

$$P_1(x) = \frac{x - x_1}{x_0 - x_1} f(x_1) + \frac{x - x_0}{x_1 - x_0} f(x_2)$$

$$R_1(x) = \frac{(x-x_0)(x-x_1)}{2!} f''(\xi(x))$$

$$\int_a^b f(x) dx = \int_a^b \left(\frac{x - x_1}{x_0 - x_1} f(x_1) + \frac{x - x_0}{x_1 - x_0} f(x_2) \right) dx + \int_a^b \left(\frac{(x-x_0)(x-x_1)}{2!} f''(\xi(x)) \right) dx$$

Example

$$\int_0^2 \sin x \, dx = ? \quad \text{by using Trapez Method}$$

$$I = \frac{h}{2} [f(x_0) + f(x_1)] - \frac{h^3}{12} f''(\xi(x))$$

$$f(x) = \sin x \quad f(0) = \sin 0 = 0 \quad f(2) = \sin 2 = 0.909$$

$$I = \frac{2-0}{2} (0 + 0.909) + \frac{2}{3} \sin(\xi) \cong 0.909$$

$$\text{Error} = \left| \frac{2}{3} \sin(\xi) \right| \leq 0.606$$

Actual solution is 1.416

Example

$$\int_0^2 (x+1) dx = \frac{h}{2} [f(x_0) + f(x_1)] - \frac{h^3}{12} f''(\xi(x))$$

$$= \frac{2}{2} (1 + 3) - \frac{8}{12} * 0 = 4 \quad \text{with zero (no) error}$$

Actual solution is also 4

$$\int_0^2 \sin x \, dx = ? \quad \text{by using Simpson Method}$$

$$I = \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)] - \frac{h^5}{90} f''''(\xi(x))$$

$$h = \frac{2-0}{2} = 1, \quad f(x) = \sin x \quad f(0) = \sin 0 = 0 \quad f(1) = \sin 1 = 0.841 \quad f(2) = \sin 2 = 0.909$$

$$I = \frac{1}{3} (0 + 4 * 0.841 + 0.909) + \frac{1}{90} \sin(\xi) \cong 1.424$$

$$\text{Error} = \left| \frac{1}{90} \sin(\xi) \right| \leq 0.909/90 = 0.01$$

Actual solution is 1.416

Example

$$\int_0^2 x^3 dx = \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)] - \frac{h^5}{90} f''''(\xi(x))$$

$$I = \frac{1}{3} [0 + 4 + 8] - \frac{1}{90} f''''(\xi(x)) = 4$$

Actual solution is also 4